Advanced Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Composite functions with restrictions

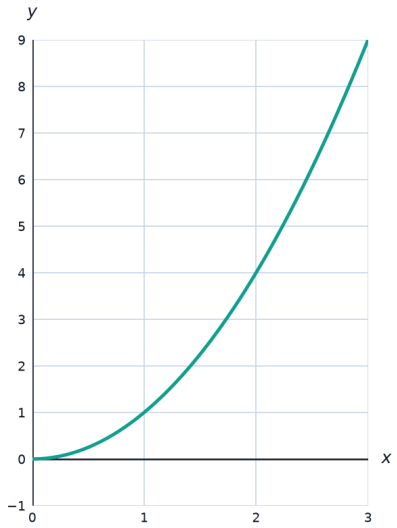
If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

2 [Foundational] • Target: Composite functions with restrictions

If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

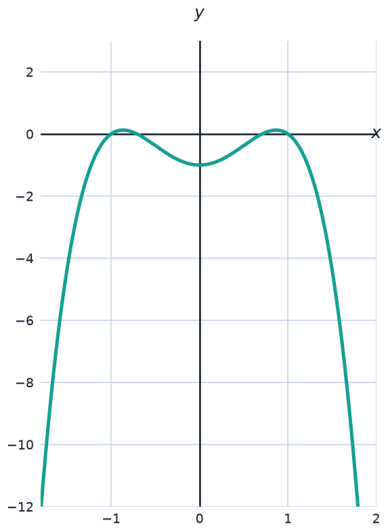
3 [Foundational] • Target: Inverse functions with domain restrictions

Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.



4 [Foundational] • Target: End behavior of polynomial functions

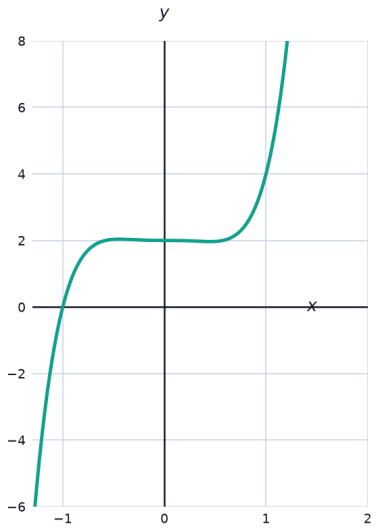
Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.



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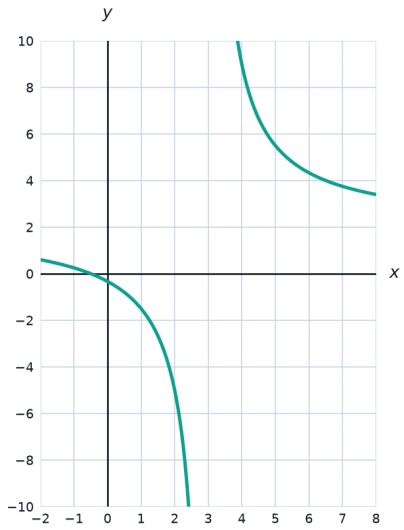
5 [Foundational] • Target: End behavior of polynomial functions

Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.



6 [Foundational] • Target: Rational functions: asymptotes and holes

Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.



7 [Foundational] • Target: Rational functions: asymptotes and holes

Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2-4}{x-2}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

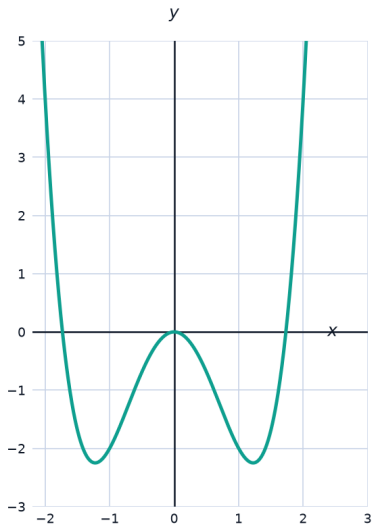
8 [Intermediate] • Target: Logarithmic functions and their properties

Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.

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9 [Intermediate] • Target: Even, odd, and periodic function classification

Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.



10 [Intermediate] • Target: Piecewise-defined functions and continuity

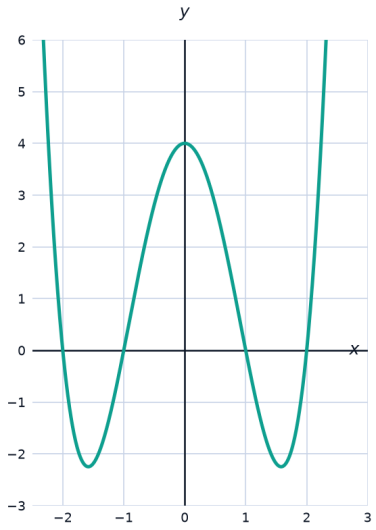
A function is defined as $f(x) = 3x + 1$ for $x < 2$ and $f(x) = kx - 3$ for $x \geq 2$. Find the value of k that makes f continuous at $x = 2$.

11 [Intermediate] • Target: A multi-part function transformation and composition problem

This question has two parts. Given $f(x) = 2x - 3$ and $g(x) = x^2$. (a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, in simplified form. (b) Find the value of x for which $(f \circ g)(x) = (g \circ f)(x)$.

12 [Intermediate] • Target: Finding zeros of higher-degree functions graphically and algebraically

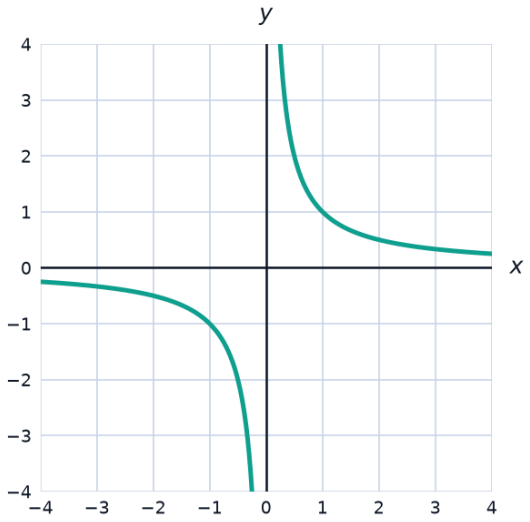
A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.



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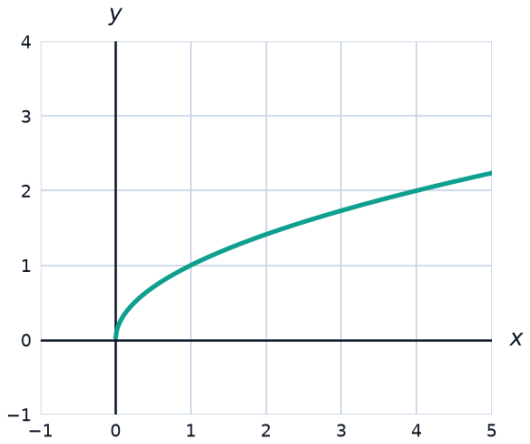
13 [Intermediate] • Target: Symmetry and transformations of rational functions

Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.



14 [Intermediate] • Target: Finding a function's formula from a description of transformations

The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.

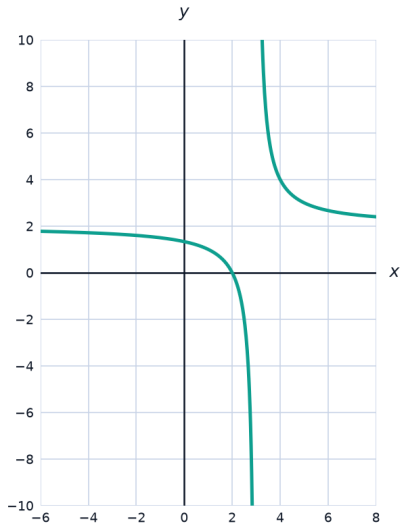


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: A multi-part rational function analysis problem • Cross-Lesson: 4.01 Advanced Trigonometry

This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.

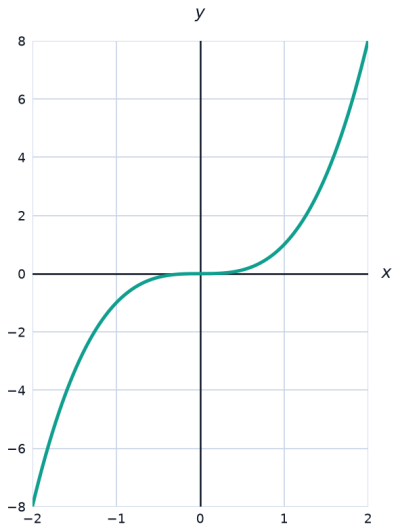


16 [Challenging] • Target: Finding a quadratic function from a table of values • Cross-Lesson: 5.01 Pre-Calculus Concepts

A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a, b, and c.

17 [Challenging] • Target: Determining if a function has an inverse using the horizontal line test • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.



18 [Challenging] • Target: End behavior of polynomial functions • Cross-Lesson: 7.01 Advanced Plane Geometry

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

19 [Challenging] • Target: Rational functions: asymptotes and holes • Cross-Lesson: 8.01 Statistics, Counting & Probability

Determine whether the function $g(x) = \frac{x^3}{x^2 + 4}$ is even, odd, or neither.

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20 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 1.01 Advanced Number Theory

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Composite functions with restrictions

Describe the end behavior of $f(x) = 4x^4 - x^3$: as $x \to \infty$, $f(x) \to$?, and as $x \to -\infty$, $f(x) \to$?

a. $-\infty, \infty$

b. $\infty, -\infty$

c. $-\infty, -\infty$

d. ∞, ∞

22 [Foundational] • Target: Inverse functions with domain restrictions

Find the vertical asymptote of $f(x) = \frac{x-15}{x+14}$.

a. -13

b. -15

c. 14

d. -14

23 [Foundational] • Target: End behavior of polynomial functions

The quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

a. $-2, -1, 1, 2$

b. $-1, -2, 1, 2$

c. $1, 2$

d. $-2, 2$

24 [Foundational] • Target: Rational functions: asymptotes and holes

The function f is defined by $f(x) = \frac{ax+8}{cx+d}$ for all real numbers $xeq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. 10

b. 16

c. -2

d. 2

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25 [Foundational] • Target: Logarithmic functions and their properties

A function f is defined for all real numbers x such that $xeq0$ and $xeq1$. If $f(x) + 2f(\frac{1}{1-x}) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

26 [Foundational] • Target: Even, odd, and periodic function classification

A piecewise function is defined by $f(x) = 4x + 2$ for $x < -3$, and $f(x) = 3x - 4$ for $x \geq -3$. Find $f(-4)$.

a. -15

b. -17

c. -14

d. -13

27 [Foundational] • Target: Piecewise-defined functions and continuity

A polynomial of even degree with positive leading coefficient has end behavior:

a. $y \to -\infty$ as $x \to \pm\infty$

b. $y \to \infty$ as $x \to +\infty$ and $y \to -\infty$ as $x \to -\infty$

c. $y \to +\infty$ as $x \to \pm\infty$

d. Constant

28 [Foundational] • Target: A multi-part function transformation and composition problem

A function has zeros at $x = -1, 0, 2$ and positive leading coefficient. Sketch shows which end behavior?

a. Left: $f \to +\infty$; Right: $f \to -\infty$

b. Left: $f \to -\infty$; Right: $f \to +\infty$

c. Both ends to $+\infty$

d. Both ends to $-\infty$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Finding zeros of higher-degree functions graphically and algebraically**

The function f is defined for all real numbers $xeq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7

30 **[Intermediate]** • **Target: Function operations: sum, difference, product, quotient**

The function f is defined by $f(x) = -x^2 + bx + c$, where b and c are constants. In the standard (x, y) coordinate plane, the maximum value of $f(x)$ is 14, and $f(0) = 5$. If $b > 0$, what is the value of $f(5)$?

a. 5

b. 10

c. 13

d. 14

31 **[Intermediate]** • **Target: Symmetry and transformations of rational functions**

Consider the function $f(x) = \frac{x^3-8}{x^2-4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

32 **[Intermediate]** • **Target: Finding a function's formula from a description of transformations**

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation $f(x) + f\left(\frac{x-1}{x}\right) = 1 + x$. What is the value of $f(2)$?

a. $\frac{3}{4}$

b. $\frac{5}{4}$

c. $\frac{7}{4}$

d. $\frac{9}{4}$

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33 **[Intermediate]** • **Target: A multi-part rational function analysis problem**

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2

c. 3

d. 4

34 **[Intermediate]** • **Target: Finding a quadratic function from a table of values**

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

35 **[Intermediate]** • **Target: Determining if a function has an inverse using the horizontal line test**

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. 20

b. 24

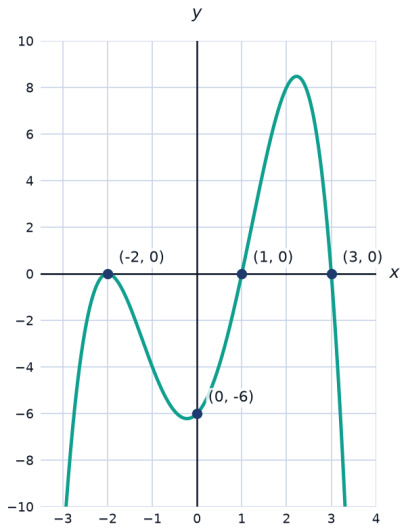
c. 32

d. 44

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36 **[Intermediate]** • **Target: Composite functions with restrictions**

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?



a. 24

b. 18

c. 17

d. 14

37 **[Intermediate]** • **Target: Inverse functions with domain restrictions**

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?

a. $8 - 2\sqrt{6}$

b. 4

c. $\frac{10}{3}$

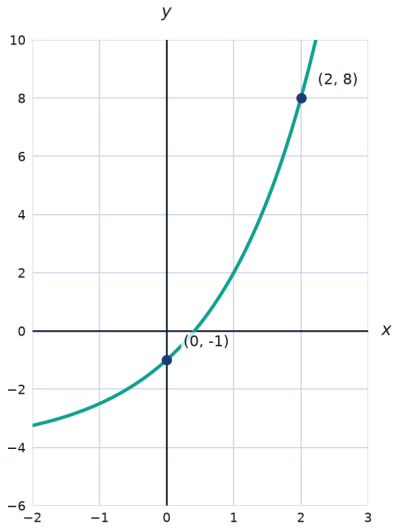
d. $2\sqrt{6}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: End behavior of polynomial functions • Cross-Lesson: 7.01 Advanced Plane Geometry

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?



a. $-\frac{5}{2}$

b. $-\frac{7}{2}$

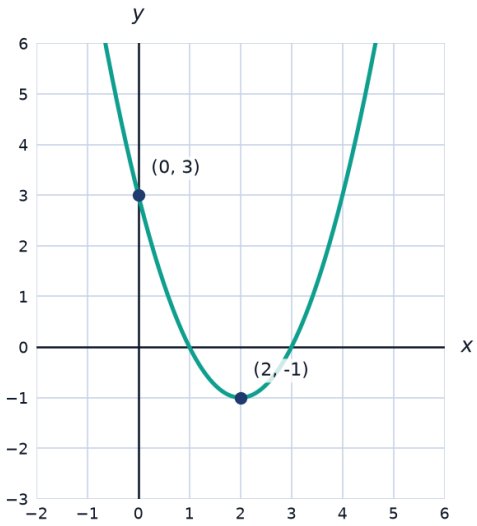
c. $-\frac{13}{4}$

d. $\frac{3}{2}$

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39 [Challenging] • Target: Rational functions: asymptotes and holes • Cross-Lesson: 8.01 Statistics, Counting & Probability

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$. The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?



a. -5

b. -3

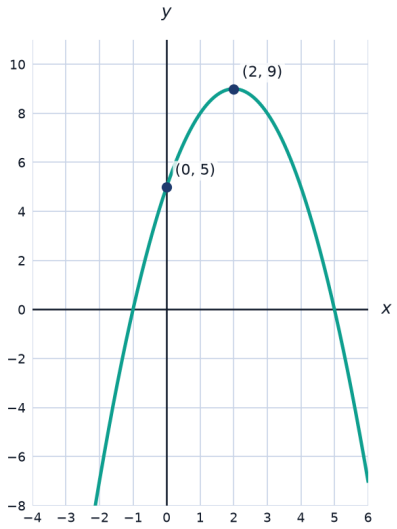
c. 1

d. 5

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40 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 1.01 Advanced Number Theory

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?



a. -4

b. -2

c. -1

d. 1

41 [Challenging] • Target: Even, odd, and periodic function classification • Cross-Lesson: 2.01 Advanced Algebra & Systems

The function f is defined by $f(x) = \frac{2x^2 + kx - 6}{x - 2}$ for all real numbers $xeq2$, where k is a constant. If the graph of $y = f(x)$ in the standard (x, y) coordinate plane has a removable discontinuity (a hole) at the point $(2, L)$, what is the value of L?

a. -1

b. 3

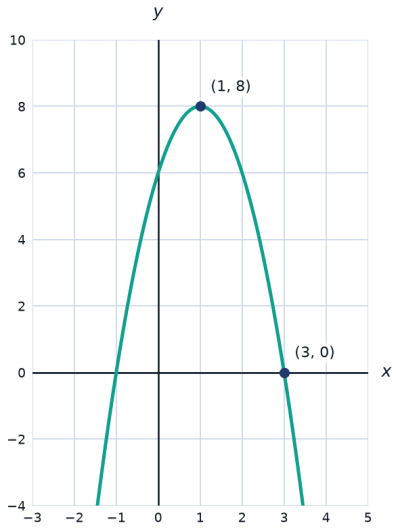
c. 7

d. 11

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42 **[Challenging]** • **Target: Piecewise-defined functions and continuity** • **Cross-Lesson: 4.01 Advanced Trigonometry**

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?



a. -10

b. -6

c. 6

d. 10

43 **[Challenging]** • **Target: A multi-part function transformation and composition problem** • **Cross-Lesson: 5.01 Pre-Calculus Concepts**

The functions f and g are defined for all real numbers by $f(x) = 2x^2 - 3$ and $g(x) = 3x + 1$ What is the value of $f(g(-2))$?

a. -53

b. 16

c. 47

d. 95

44 **[Challenging]** • **Target: Finding zeros of higher-degree functions graphically and algebraically** • **Cross-Lesson: 6.01 Advanced Coordinate Geometry**

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

a. -2

b. 2

c. 7

d. 16

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45 [Challenging] • Target: Function operations: sum, difference, product, quotient • Cross-Lesson: 7.01 Advanced Plane Geometry

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

a. 4

b. 3

c. 1

d. 6

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